

TROY WU

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EDUCATION

University of British Columbia

Bachelor of Computer Science and Business | GPA: 3.85/4.00

Expected Graduation: December 2027

Vancouver, BC

- Dean's Honour Roll, Trek Excellence Scholarship (Top 5%), Ranked 6/791 in UBC Sauder.
- Courses:** Data Structures & Algorithms; OOP (Java); Operating Systems; Relational Databases.

EXPERIENCE

Tesla

August 2026 - December 2026

Incoming Software Engineer Intern - Robotaxi & Grok Validation

Palo Alto, CA

Amazon

May 2026 - August 2026

Software Development Engineer Intern - Everyday Essentials

Vancouver, BC

- Served as the **first engineer in a team of applied scientists**, automating and productionizing machine learning infrastructure from research prototype to scaled production pipeline.
- Designed and built an end-to-end Python ML inference pipeline scoring product relationships across **80M+ products**, powering recommendations attributed to **\$330M+ annualized sales** across Amazon NA storefronts.
- Reduced batch inference time from **4 days to 8 hours** by orchestrating 80+ parallel GPU jobs on g5.4xlarge instances with **Airflow and AWS Batch**, with built-in fault tolerance and shard-level resumability.
- Designed a S3 feature store serving **1TB of product embeddings** via parallel batch lookups, eliminating a 90-minute fetch bottleneck and cutting per-run data costs from **\$13 to under \$1**.

D2L (Desire2Learn)

January 2026 - April 2026

Software Engineer Intern - Lumi (Brightspace GenAI)

Vancouver, BC

- Developed full-stack GenAI features for Brightspace LMS deployed across **400+ institutions** and **260K+ annual generations** using Web Components/LitElement, C# MVC, and SQL databases.
- Built Brightspace's **AI Audio Overview** feature end-to-end, delivering natural AI-narrated content summaries via serverless **Claude Sonnet + AWS Bedrock/Lambda and Polly TTS** pipeline to support diverse learning styles.
- Built and refactored Playwright E2E tests for critical GenAI workflows **expanding test coverage by 28%**, and consolidated duplicate code through shared TestBench fixtures **reducing suite runtime by 32%**.

Absolute Security

May 2024 - December 2024

Software Engineer Intern

Vancouver, BC

- Built and deployed full-stack, telemetry security services (Java, JavaScript/Manifest V3 extension) for 1M+ enterprise Chromebook endpoint devices, **enabling continuous monitoring and threat detection**.
- Engineered a scalable Java microservice integrated with **AWS Grafana** to automate and standardize log exports for 1M+ devices, **boosting error detection speed by 30%** and QA efficiency.
- Rebuilt QA log interface (HTML, CSS, JS), **reducing export latency by 50%** and improving usability.

PROJECTS

BasketBot – ProduHacks Winner 🏆 | Python, OpenCV, TensorFlow, PyAudio, VOSK

- Winner of ProduHacks (50 teams, \$400) by building an object-detection-based automatic basketball scorekeeper.
- Implemented TensorFlow-based machine learning models to recognize basketball nets for accurate scoring.

Campus Explorer 🗺️ | TypeScript, Node.js, Express.js, JavaScript, HTML/CSS, Google Maps API

- Built full-stack web application with TypeScript/Express.js backend and interactive JavaScript frontend enabling real-time querying and visualization of 300+ campus rooms via Google Maps API.

LEADERSHIP

BUCS Community Executive | UBC Business and Computer Science Major Community Team

- Increased YoY engagement by 31% for 500+ students in BUCS by planning & organizing community events.

TECHNICAL SKILLS

Languages: Python, Java, C#, TypeScript, JavaScript, C++, SQL.

Frameworks/Libraries: PyTorch, TensorFlow, pandas, NumPy, React.js, Node.js, Express.js, Flask, OpenCV, JUnit.

Developer Tools: AWS (Batch, DynamoDB, Bedrock, Lambda, S3), Airflow Docker, Kubernetes, Databricks, Git, MongoDB.